

Data Profiling Report

Findings from the Clinical Trials Dashboard

1. Overall clinical trial volume

The dataset contains approximately **8,089 clinical trials** conducted across **133 countries**. This indicates that the dataset provides broad international coverage of clinical research activity.

2. Trial status

The two largest groups are **Completed** and **Recruiting** trials.

- Completed trials: approximately **2,659**
- Recruiting trials: approximately **2,563**

This shows that a substantial proportion of the trials have either already finished or are currently recruiting participants. Other statuses, such as active but not recruiting, not yet recruiting, suspended, and terminated, represent smaller portions of the dataset.

3. Geographic distribution

The dashboard shows that clinical trials are **not evenly distributed across countries**. A relatively small number of countries account for a large share of the trials.

The **United States** has the highest number of trials among the countries shown in the Top 10 visualization, followed by countries such as **France, the Netherlands, Turkey, the United Kingdom, Italy, China, Spain, Egypt, and Canada**.

This suggests that clinical research activity is concentrated in particular countries rather than being evenly distributed worldwide.

4. Age eligibility

The age analysis shows that **Adult** participants make up the largest age category, accounting for approximately **74.3%** in the displayed visualization.

Other categories, including child/adolescent and older-adult groups, represent substantially smaller shares. This indicates that the dataset contains considerably more trials involving adult populations than pediatric populations.

5. Gender eligibility

Gender eligibility is strongly dominated by the **All** category, which accounts for approximately **96.82%** of the displayed trials.

Female-only and male-only trials represent much smaller proportions. This suggests that most trials in the dataset are designed to allow participation from **both genders** rather than being restricted to one gender.

6. Trial activity over time

The time-trend visualization shows a **very large increase in clinical-trial activity around 2020**, followed by a substantial decline in subsequent years.

The dashboard therefore suggests that trial activity was particularly high around **2020–2021**, compared with the years before and after this period.

Important: The sharp decline after 2021 should be interpreted carefully because later years may contain incomplete or differently represented records in the dataset.

7. Study type

The earlier analysis showed:

- **Interventional:** 4,586 trials
- **Observational:** 3,509 trials

Therefore, **interventional studies are more common** than observational studies in the dataset.

8. Participant enrollment

The enrollment analysis shows considerable variation in the number of participants across trials. Most trials have substantially smaller enrollment than a few extremely large studies.

The earlier QA identified **24 very-high-enrollment records**, so extreme values should be considered when interpreting averages and the enrollment distribution.

Overall, the clinical-trial dataset contains approximately 8,089 trials across 133 countries. Clinical research is geographically concentrated, with the United States contributing the largest number of trials among the leading countries. Completed and recruiting trials are the two dominant study-status categories. Adult participants are the most common target population, while most trials are open to all genders. Interventional studies are more common than observational studies. Trial activity shows a pronounced increase around 2020–2021, followed by a decline in later years, although the later-year pattern should be interpreted cautiously.

QA SUMMARY

QA Check	Result	Interpretation
Missing result_first_posted	669 (8.23%)	Some records do not contain a first-posted result date.
Missing interventions	1,225 (15.07%)	Interventions are missing for a noticeable proportion of trials.
Missing outcome_measures	37 (0.46%)	Only a small proportion of trials have missing outcome measures.
Very high enrollment	24 records	These records require review because their enrollment values are unusually large.

Other cleaning/validation performed

- Date fields were standardized to a consistent **DD-MM-YYYY** format.
- Missing **gender** values were replaced with **All**.
- Missing **country** values were replaced with **Other**.
- Country information was extracted from the location data.
- age_category was created from the age information.
- Missing age_category values were handled using the available age information.
- Empty/missing **outcome measure** records were removed as part of cleaning.
- Missing **study design** records were removed.
- The study_documents column was removed.
- The cleaned data was validated in **Jupyter Notebook** and used for the Power BI dashboard.

Overall QA conclusion

The data-cleaning and QA process improved the consistency and usability of the clinical-trial dataset. Most important fields were retained and standardized, while missing values and inconsistent information were addressed where appropriate. The main remaining data-quality concerns are missing interventions (15.07%), missing first-posted results (8.23%), and 24 unusually high enrollment records that require further review. Therefore, the cleaned dataset is suitable for exploratory analysis and Power BI visualization, while the identified QA issues should be considered when interpreting the results.

Recommendations

1. Improve completeness of intervention data

Since **15.07% of records have missing intervention information**, future data collection should make intervention details mandatory where applicable. This would improve the ability to compare different treatments and interventions.

2. Improve reporting of results

Approximately **8.23% of records are missing result_first_posted**. Trial records should be updated promptly when results become available so that the dataset provides a more complete view of trial outcomes.

3. Review unusually high enrollment values

The QA process identified **24 records with very high enrollment values**. These records should be manually reviewed to determine whether the values are genuine or caused by data-entry or formatting issues.

4. Standardize data entry

Dates, countries, gender, age categories, and other categorical fields should follow predefined formats and controlled values. This reduces inconsistencies and makes future analysis easier.

5. Improve country/location information

Location fields should use standardized country names rather than relying only on free-text locations. This would improve geographic analysis and reduce difficulties when extracting countries from location descriptions.

6. Maintain regular data-quality checks

Automated QA checks should be performed whenever new trial data is added. Checks should include:

- Missing values
- Duplicate trial IDs
- Invalid dates
- Unusual enrollment values
- Invalid categorical values
- Inconsistent country names

7. Use the Power BI dashboard for monitoring

The dashboard can be updated regularly to monitor:

- Trial status
- Recruiting vs completed trials
- Geographic distribution
- Age and gender eligibility
- Study types and designs
- Trial activity over time

Overall recommendation

The cleaned clinical-trial dataset should be maintained through standardized data-entry rules and regular automated quality checks. Particular attention should be given to missing intervention and result-posting information and to unusually high enrollment values. Continued use of the Power BI dashboard would support easier monitoring, identification of trends, and data-driven decision-making.

Conclusion

This project successfully transformed the raw clinical-trial dataset into a cleaner and more structured dataset suitable for analysis and visualization. The data-cleaning process included standardizing date formats, handling missing values, removing unnecessary fields and incomplete records where appropriate, and creating additional variables such as **country** and **age_category**. Quality assurance checks were also performed to identify missing information, inconsistencies, and unusually high enrollment values.

The analysis of approximately 8,089 clinical trials across 133 countries showed that clinical research activity is concentrated in a relatively small number of countries. Completed and recruiting trials were the largest study-status groups, while adult participants represented the largest age category and most trials were open to all genders. Interventional studies were also more common than observational studies. The Power BI dashboard provided an interactive way to explore these patterns and examine trial activity across status, geography, age, gender, study type, study design, enrollment, and time.

The QA process identified areas that require further attention, particularly missing intervention information, missing result-posting dates, and 24 unusually high enrollment records requiring review. Overall, the cleaned dataset provides a useful foundation for exploratory analysis and reporting. Future improvements should focus on increasing data completeness, standardizing data-entry practices, conducting regular automated quality checks, and maintaining the Power BI dashboard for ongoing monitoring and decision-making.